

Assessment Brief

Module title:	Cyber Security and Applied Cryptography	Module code:	CSY3062
Assessment code/title:	Security report with recommendations for delivering data security	Assessment weighting/word-limit:	75%
Submission date:	6 th April 2026	Feedback date:	4 weeks after the submission date
Module Leader:	Prakash Chandra Adhikari (prakash.adhikari@nami.edu.np)	Resit date:	TBA
Assessment Task:			

Title: Security report with recommendations for delivering data security

Setting the scene: Please, read the full details below- Assignment Scenario.

Learning Outcomes aligned to this Assessment:

Learning Outcomes are available on the Module Specification for the module, and are also on the Nile site

Academic Practice (referencing style, literature usage, AI Usage):

Please, read the assessment brief below.

Use of Generative AI (Artificial Intelligence) within this Assessment:

Some uses of Generative AI may be deemed unethical in your Assessment.

[Please access the following position guidance from University of Northampton on the use of Generative AI within assessments.](#)

This assignment falls into the following category:

Category 2: GenAI can be used in an assistive role

You may make use of GenAI in your assessment in an assistive role, but you must acknowledge this appropriately.

Ref to the UON guidance on the different types of GenAI use, what you need to acknowledge and how you should do this.

Assessment Guidance:

Reading List: You will find a link to your online reading list on NILE in the “About this module” section.

Include skills hub link:

You may refer to the following link: <https://skillshub.northampton.ac.uk/#>

Confidential advice available through su.advice@nothampton.ac.uk

Feedback:

Feedback should be received within 20 working days

Announcement will be sent out on Nile

Instruction on Anonymity for students [Further guidance is available online](#)

Assignment Scenario

Gradgo the Graduate Recruitment Agency in Northampton have considered your responses to their initial Cyber Security queries and have now decided to extend your commission for a comprehensive report in the form of a term (research) paper.

You are asked to write a security report in the form of a term paper, in which you propose a suitable “data security” solution for any type of software-based activity you may consider that the Agency would be engaged on as part of their total business operations on the Internet.

In your paper you should describe your chosen area in detail. Describe any identifiable vulnerabilities- conduct a vulnerability test. Explain what you would recommend and consider to be a suitable mitigation solution. Identify and experiment with suitable cryptographic software tools. Explain how you would implement your recommended solution, based on your own findings, to provide data security for the Agency.

Note:

- You are a Cyber Security professional with expertise in applied Cryptography. Cryptography should be the main basis of your solution and recommendations.
- You may choose* any online application, or a security technology that might provide data security for online applications.
- Your paper should be clear, concise, and written in a technical context.
- You will be expected to present your findings to an audience of other security experts.
- You may discuss the topic of your choice with your tutor during the assignment support session.

*There are several topics listed below for your reference.

Your assignment tasks in a nutshell.

1. Research, investigate, and evaluate the security implications of an **online Application**.

Or

Conduct an in-depth analysis of a **Security Technology** that can provide online Security when deployed.

2. Choose and experiment with suitable **cryptographic software tool(s)***,

Or

Develop your own software tool that can implement any form of cryptography,

which could potentially be used as part of the provision of **data security** within the context of your chosen topic.

3. Write a full security report in the form of a 2000-4000 words term paper. Appendices are not considered in the word count.
4. Present your paper in the form of a video **.

*. Open source & free

** The purpose of the presentation is twofold.

a) Share your assignment practical experimentation with your tutor

b) Demonstrate your awareness and knowledge of your chosen topic for assessment purposes.

A typical technical term paper structure is as follows.

Title

Authors details

Abstract

Introduction: background information, related work, major issues, and scope of the investigation.

Main Body: your approach to address the issue(s) of your choice, your findings, review-comparative analysis- detailed technical underpinning, security software review, use & analysis*, development of solutions etc. You can break this part into several sections.

Summary: conclusions, recommendations- your contribution in general.

References: books, journals, papers, websites, etc.

*This would typically, involve some form of a practical experimentation.

Research Paper possible topics

- Cyber Security issues*
- Penetration testing & Ethical Hacking for improved security
- Practical Cryptographic systems
 - Security Standards,
 - Encryption algorithms, Hash functions with appendix and message recovery models,
 - Authentication Systems,
 - PKI
- Secure Protocols: IPSec, SSL/TLS
- Quantum resistant cryptography
- WiFi technologies
- Mobile technologies
- Networking technologies: VPNs
- E-commerce Security
- Systems Security- Windows/ Unix/Linux cryptographic schemes

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Identity and access Management

Social engineering

Insider Threat

MDM/Legacy/IOT security

Supply chain Vulnerabilities

SIEM Integration

Sophisticated Threat Landscape due to AI/other emerging threats

There are several online sources, to include the following, that publish technical papers and articles on various computer security topics. It is known that you may be required to register to download some of their specific documents. In such cases, you can register as a University of Northampton student and give your course details.

<http://www.techrepublic.com/topic/security/>

Grading:

The full grading criteria table is available to view on Nile.

Grading criteria will be based on taking the following into consideration.

1. The level of challenge met by the student.
2. Knowledgeable and logical adherence to issues.
3. Evidence of clearly focused and relevant research effort.
4. Effective, appropriate, and clear choice and use of S/W tool.
5. Written with appropriate technical detail and presented with clarity and creativity.

Assessment Submission

To submit your work electronically, please go to the 'Submission' area on the NILE site and use the relevant submission points to upload the assignment deliverables. The deadline for this is 23:59 (UK local time) on the date of submission. Please note that Essays and text-based reports should be submitted as Microsoft Word documents (.doc or .docx), or as guided within the assignment. [Please access the following guide to submitting assessments.](#)

Written work submitted to [Turnitin](#) will be subject to anti-plagiarism detection software. Turnitin checks student work for possible textual matches against internet available resources and its own proprietary database. [Please access the University of Northampton's Plagiarism Avoidance Course \(UNPAC\) to learn more.](#)

When you upload your work correctly to Turnitin you will receive a receipt which is your record and proof of submission. If your assessment is not submitted to Turnitin, rather than a receipt, you will see a green banner at the top of the screen that denotes successful submission.

Further Assessment Guidance:

Please access the following document for more general information about the assessment process, including anonymous marking, submissions, and where to find feedback and grades.

[UON Standard Assessment Guidance.](#)

N.B Work emailed directly to your tutor will not be marked.